Logarithmic scales

1 A log scale relating x and y is shown in the table of values below:

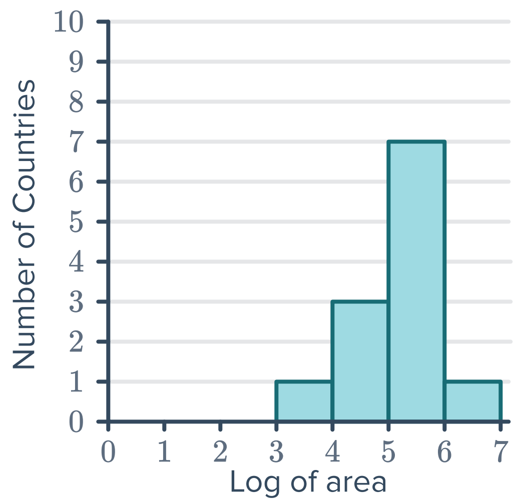
Log scale measure (y)		Linear measure (x)
0	$=$	1
1	$=$	10
2	$=$	100
3	$=$	1000
4	$=$	$10\,000$

Form an equation relating x and y. Express the equation in logarithmic form.

2 The given histogram shows the area (in square kilometers) for 12 countries using a $\log_{10}$ scale on the x-axis:

a How many countries have an area of between 10^5 and 10^6 square kilometers?

b How many countries have an area of between $10\,000 \text{ km}^2$ and $100\,000 \text{ km}^2$?



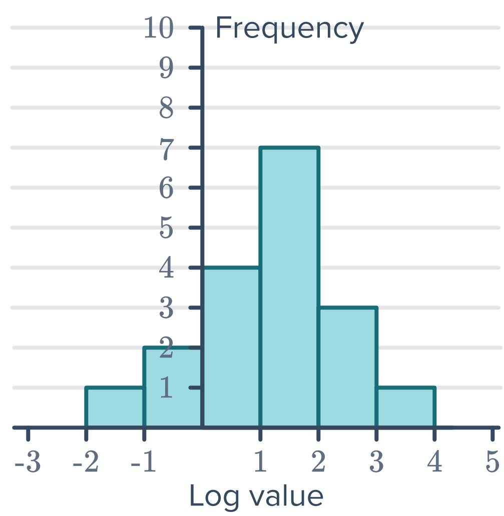
3 The histogram has been plotted on a $\log_{10}$ scale:

a How many scores are between 10^0 and 10^1?

b What is the total frequency of scores between 1 and 100?

c What is the total frequency of scores larger than 0.1?

d What is the total frequency of scores smaller than 10?



4 The histogram shows the masses (in grams) [redacted]roup of insects and animals, plotted on a $\log_{10}$ scale:

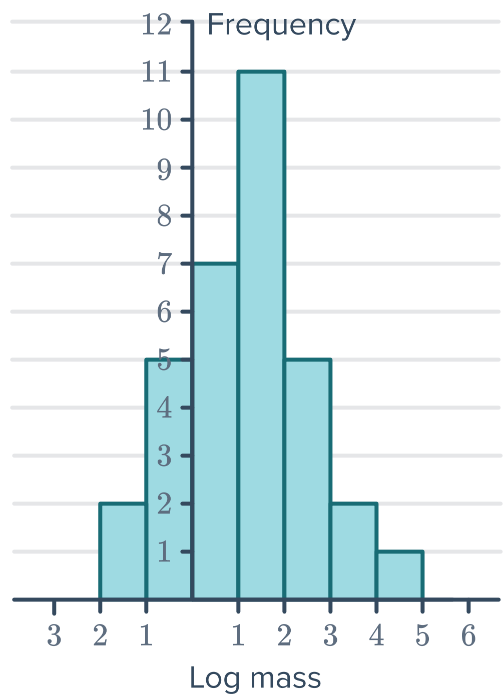
a How many animals and insects have a mass between 10^1 and 10^2 grams?

b How many animals and insects are included in the data altogether?

c A caterpillar has a mass of 3 grams. What value does this take on a $\log_{10}$ scale? Round your answer to two decimal places.

d Which interval on the histogram would include a cat that has a mass of $4.5\ \mathrm{kg}$?

e From the data in the histogram, how many of the animals or insects had a mass that was over 100 grams?



5 A financial planner has clients with a large range of annual incomes. To manage the data more easily, the income (in dollars) of each client is first converted to a $\log_{10}$ scale.

a What income would result in 4.9 on a $\log_{10}$ scale?

b What range of incomes corresponds to a range of 5 to 6 on the $\log_{10}$ scale?

6 The lifespans of various animals and insects [redacted]hown in the table below:

Animal/Insect	Lifespan (days)	Lifespan in a $\log_{10}$ scale
Ant	20	
African elephant	25 500	4.41
Bowhead whale	71 400	
Bumblebee	21	1.32
Chicken	2920	3.47
Dog	5110	
Emperor penguin	8500	3.93
Fly	18	
Giraffe	10 120	
Hamster	780	2.89
Ladybug	365	
Little penguin	2190	3.34
Monkey	7300	
Shark	10 980	4.04

a Convert the lifespans to a $\log_{10}$ scale to complete the table, rounding each value to two decimal places. Some values have already been filled in.

b Create a histogram to represent this data, on a $\log_{10}$ scale.

Applications of logarithms

7 The number of registered nurses working in hospitals t years after the year 2002 can be modeled by the equation $N = 28\log_4(t+2)$, where N represents the number of nurses in thousands.

a How many registered nurses were working in hospitals in the year 2002?

b Find number of registered nurses in 2004.

8 A major communications company found that the more they spend on advertizing, the greater their revenue. Their sales revenue R, where x represents the amount they spend on advertizing (R and x in thousands of dollars) is given by:

$$R = 10 + 20 \log_4 (x + 1)$$

a Determine their sales revenue if they spend no money on advertizing.

b Determine their sales revenue if they spend $\$14\,000$ on advertizing. Round your answer to the nearest thousand.

c Would you say that every extra $\$1000$ spent on advertizing becomes more or less effective in terms of raising revenue?

9 The Richter scale is used to measure the intensity of earthquakes. The formula for the Richter scale rating of a quake is given by: $R = \log x - \log a$, where a is the intensity of a minimal quake that can barely be detected, and x is a multiple of the minimal quake's intensity.
The given table shows how quakes are categorized according to their Richter scale rating:

	Richter rating
Minor	$2 - 3.9$
Light	$4 - 4.9$
Moderate	$5 - 5.9$
Strong	$6 - 6.9$

a Rewrite the formula as a single lograithm.

b A seismograph measures the intensity of a quake to be $x = 5711a$.

i Find the Richter scale rating R of this quake to one decimal place.

ii In which category does the quake fall?

c A seismograph measures the intensity of an earthquake to be $15\,850$ times the intensity of a minimal quake.

i Find the Richter scale rating R of this quake to one decimal place.

ii In which category does the quake fall?

10 As elevation A (in meters) increases, atmospheric air pressure P (in pascals) decreases according to the equation

$$A = 15\,200 \, (5 - \log P)$$

Trekkers are attempting to reach the 8850 m elevation of Mt. Everest's summit. When they set up camp at night, their barometer shows a reading of $45\,611 \text{ Pa}$. How many more vertical meters do they need to ascend to reach the summit? Round your answer to the nearest meter.

11 The Richter Scale is a base 10 logarithmic scale used to measure the magnitude of an earthquake, given by $R = \log_{10} s$, where s is the relative strength of the quake. Some past earthquakes and their Richter scales are shown in the table below:

Earthquake	Richter scale
Sumatra 2004	9.1
China 2008	7.9
Haiti 2010	7.0
Italy 2009	6.3

a How many times stronger is n earthquake that registers 4.0 on the Richter scale than an earthquake that measures 3.0?

b How many times stronger is a quake of 7.6 than one of 5.2? Round your answer to the nearest whole number.

c How many times stronger was the earthquake in Sumatra compared to the earthquake in China? Round your answer to the nearest whole number.

d The aftershock of an earthquake measured 6.7 on the Richter Scale, and the main quake was 4 times stronger. Find the magnitude of the main quake on the Richter Scale, to one decimal place.

12 The sound level or loudness, L, of a noise is measured in decibels (dB), and is given by the formula: $L = 10\log\left(\frac{I}{A}\right)$, where I (in watts/cm^2) is the intensity of a particular noise and A is the the intensity of background noise that can barely be heard.

a At a concert, standing near a speaker exposes you to noise that has intensity of about $I = 0.5 \times 10^{13} A$.

- **i** How many decibels is this? Round your answer to the nearest dB.
- **ii** Noises measuring up to 85 dB are harmless without ear protection. By how many decibels does the noise at a concert exceed this safe limit? Round your answer to the nearest dB.

b The maximum intensity which the human ear can handle is about 120 dB. The noise in a recycling factory reaches 132.9 dB. How many times louder than the maximum intensity is the factory noise? Round your answer to one decimal place.

c If one person talks at a sound level of 60 dB, find the value of L which represents the decibel level of 100 people, each talking at the same intensity as that one person.

d If a sound intensity doubles, by how much does the level of sound in decibels increase?

e Given $A = 10^{-16}$ watts/cm^2, find:

- **i** The sound level of a sound with intensity $I = 10^{-5}$ watts/cm^2.
- **ii** The sound intensity of a passenger plane passing over houses prior to landing, if the engine's loudness is registered at 103 dB. Give your answer in exact form.

13 The decibel scale, which is used to record t[redacted]udness of sound, is a logarithmic scale.

- In the decibel scale, the least audible sound is assigned the value of 0.
- A sound that is 10 times louder than 0 is assigned a decibel value of 10.
- A sound 100 (10^2) times louder than 0 is assigned a decibel value of 20.

In general, an increase of 10 decibels corresponds to an increase in magnitude of 10. The table shows the decibel value for various types of noise:

Type of noise	Decibel value
Jet plane	150
Pneumatic drill	120
Industrial	110
Stereo music	100
Inside a car	90
Office	70
Household	60
Wind turbine	50
Bedroom	30
Falling leaves	20

a How many times louder than 0 is the sound of a bedroom?

b How many times louder than 0 is the sound of an office? Write your answer as a power of 10.

c If the sound of a normal speaking voice is 50 decibels, and the sound in a bus terminal is 80 decibels, how many times louder is the bus terminal compared to the speaking voice?

d How many times louder is the sound of industrial noise than the sound of a wind turbine?

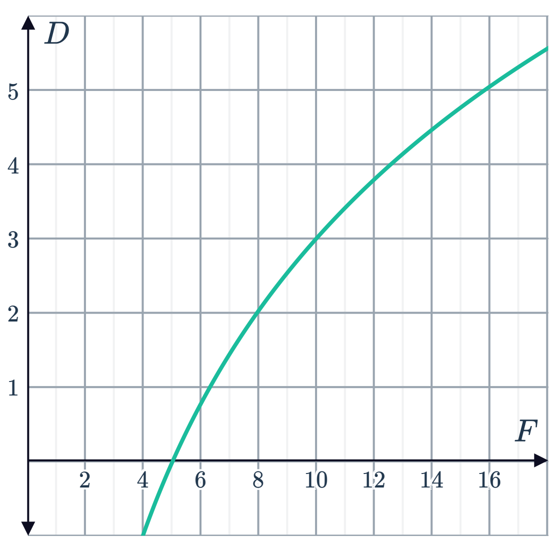
14 The signal ratio D (in decibels) of an electronic system is given by the formula: $D = 10\log\left(\frac{F}{I}\right)$, where F and I are the output and input powers of the system respectively.

a Find the input power I in megawatts if the output power is equal to 10 MW and the signal ratio is 20 decibels.

b The given graph shows the equation $D = 10\log\left(\frac{F}{5}\right)$, where the input power is 5 MW. State the interval that contains:

i The signal ratio when $F = 9$.

ii The output power when the signal ratio is $D = 1$.



15 Search engines give every web page on the Internet a score (called a Page Rank) which is a rough measure of popularity/importance. One such search engine uses a logarithmic scale so that the Page Rank is given by: $R = \log_{11} x$, where x is the number of views in the last week.

a Determine the Page Rank of a web page that received 7300 views in the last week. Round your answer to the nearest integer.

b Google uses a base-10 logarithmic scale to get a web page's Page Rank: $R = \log_{10} x$. How many more times the views did a web page with a Page Rank of 5 get, than one with a Page Rank of 3?

16 The time taken (t years) for A grams of a radioactive substance to decompose down to y grams, where k is a constant related to the properties of a particular substance is given by:

$$t = -\frac{1}{k} \log_{2.3} \left(\frac{y}{A}\right)$$

Consider the following substances, giving your answers to the nearest year:

- R_1 has a constant of $k = 0.00043$
- R_2 has a constant of $k = 0.00047$

a Find how long it takes 140 units of R_1 to decompose down to 105 units.

b Find how long it takes 200 units of R_2 to decompose down to 150 units.

c Find the half-life of each substance. That is, determine how long it takes a quantity of a substance to decompose down to half the original quantity.

d Will it take twice the half-life for each substance to decompose completely? Explain your answer.

17 Consider the function $y = 8\left(2\right)^{2x}$, for $x \geq 0$.

a The function above can be written as $\log_2 y = mx + k$. Find the values of m and k.

b Sketch the graph of $\log_2 y$ against x.

c Find the rate of change of the linear function.

d Find the value of x when $y = 16$.

18 The Palermo impact hazard scale is used to rate the potential for collision of an object near Earth. The hazard rating P is given by the equation $P = \log R$, where R represents the relative risk of collision. Two asteroids are identified as having a relative risk of collision of $\frac{6}{7}$ and $\frac{4}{5}$ respectively. Find the exact difference in their measure on the Palermo impact scale.

19 pH is a measure of how acidic or alkaline a substance is. The pH (p) of a substance can be found according to the formula: $p = -\log_{10} h$, where h is the substance's hydrogen ion concentration.
The pH scale goes from 0 to 14, with 0 being most acidic, 14 being most alkaline and pure water has a neutral pH of 7.

- **a** Store-bought apple juice has a hydrogen ion concentration of about $h = 0.0002$. Find the pH of the apple juice correct to one decimal place.
- **b** Is the apple juice acidic or alkaline?
- **c** A banana has a pH of about 8.3. Find h, its hydrogen ion concentration. Give your answer as an exact value.

20 Researchers conducted a test to determine how well information is retained through the method of rote learning. To do this, they asked students to memorize mathematical formulae in the lead up to the first test, and then study no further. They continued to test them once a month over 7 months. They found that the average student's test scores (P) decreased over time (t months), but at a slowing rate.

- **a** Which equation could be used to model their findings of the relationship between P and t under the rote learning method?

  A $P = -86 - 23\log_2(t+1)$ **B** $P = 23\log_2(t+1) - 86$ **C** $P = 86 - 23\log_2(t+1)$

  D $P = 23\log_2(t+1) + 86$

- **b** Using the equation in part (a) to model student performance over time, find the average student's test score on the initial test at $t = 0$.
- **c** Find the average student's test score on the 6th test. Round your answer to the nearest whole number.
- **d** Find the number of months it takes for the average student's test score to equal 0 using the rote learning method. Write your answer as a whole number of months.

21 The information entropy H (in bits) of a randomly generated password consisting of L characters is given by: $H = L\log_2 N$ where N is the number of possible symbols for a character in the password.
A case sensitive password consisting of seven characters is to be made using letters from the alphabet and/or numerical digits.

- **a** Find the value of L.
- **b** Find the value of N.
- **c** Find the value of H correct to two decimal places.
- **d** It was found that a seven character password resulted in an entropy of 28 bits. Find the possible number of symbols for a character.

22 Consider the formula $d = 10\log\left(\frac{P}{k}\right)$, for some constant k.

a If P is increased by a factor of 10, by how much does d increase?

b If P is increased by a factor of 5, by how much does d increase? Write your answer as an exact value.

c Determine the value of P for which $d = 0$.

Applications of exponential functions

23 The prize of a scratch ticket gives the winner two options:

- Option 1: Receive one payment of $\$500\,000$ immediately, or
- Option 2: Receive $\$1$ on the first day of winnings, double that ($\$2$) on the second day, and so on for three weeks.

a For the second option find the day on which the amount received will first exceed the value of the first option.

b How much more money does the winner receive if they take the second option?

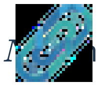
24 Under certain climatic conditions the proportion, P, of the current blue-green algae population compared to the initial population satisfies the equation $P = e^{0.007t}$, where t is measured in days from when measurement began.
Solve for t, the number of days it takes the initial number of algae to double to the nearest two decimal places.

25 The voltage V in volts across an electrical component where A is the initial voltage and t is the time in years, decays according to the equation:

$$V = Ae^{-\frac{1}{2}t}$$

Find the value of t such that V is half of the initial voltage. Round your answer to two decimal places.

26 The spread of a virus through a city where N is the number of people infected by the virus after t days, is modeled by the function:



$$N = \frac{15\,000}{1 + 100e^{-0.5t}}$$

a Calculate the number of people that will have been infected after 3 days.

b How many whole days will it take for at least 4000 people to be infected with the virus?

27 The remains of a human body can be dated by measuring the proportion of radiocarbon in tooth enamel. The proportion of radiocarbon, A, remaining t years after a human passes away is given by:

$$A = Ce^{-kt}$$

where k is a positive constant.

a Solve for the exact value of k if the amount of radiocarbon present is halved every 5594 years.

b For a particular corpse, state the number of years since the person passed away, if the amount of radiocarbon present is exactly 25% of the original amount at death.